

connected in parallel, it is necessary to ensure protection of PV panels and the system against reverse current and over-current (OC).

Fuses are primarily used to protect the system against short-circuit and fire hazards. Fuses in PV installations are subjected to extreme conditions of the environment like exposure to sunlight that causes abnormal temperature of the fuse, which affects the performance of the fuse. Thus, sizing and selection of the right fuses and cables must be ensured.

Moreover, PV modules also produce continuous current. This makes it all the more important to have proper fuse sizing.

A fuse is an over-current device that is designed to sacrifice itself to protect electrical systems. Fuses are designed to open circuits when put under stress by excessive current flow caused by overloads or faults. Choosing the right fuse for an application will prevent fires and other damages

that may be caused when something goes wrong. Typical problems may include a cable coming loose in the inverter circuit, a cable shorting to Earth, accidental cutting of a cable, an animal or rodent chewing through cabling, weather damage and so on.

Fig. 1 illustrates a typical solar system from PV modules to inverter.

Fuse ratings

Fuses are rated by current and voltage, and are usually rated solely for AC, solely for DC or for both AC and DC. If incorrect fuses are used for DC applications, voltage rating may need to be derated and you would have to consult the fuse manufacturer for further information on their product. This is because of the greater arc energy that needs to be absorbed during the breaking process.

Key steps of fuse sizing for solar PV fuse selection

Following steps should be used for

the proper sizing of a fuse of a string as per article 690.8 of National Electrical Code 2011.

1. Calculate maximum circuit current.
2. Calculate nominal fuse ampere rating.
3. De-rate fuse nominal rating due to abnormal temperature conditions.
4. Calculate the nameplate ampere rating of fuse.
5. Verify fuse rating with respect to conductor rating.

Over-current protection (either fuses or breakers) must be included in a PV source or output circuit only if you have three or more array strings. PV module manufacturers usually specify the maximum number of strings that can be connected in parallel without adding any fuse protection, by means of maximum reverse current capacity of a PV module.

Fuses are usually placed inside a combiner box (if your system is using a combiner box) or inside the DC dis-

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